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# Olfactory-Interactive 4D materials from a carvone-based photocurable resin

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**ABSTRACT**

The development of 4D materials has expanded adaptive manufacturing; however, most systems are limited to passive shape changes and lack interactive sensory functions. Here, we present an olfactory-responsive 4D material with integrated antimicrobial activity. The material exhibits shape-memory behaviour under external stimuli and releases aromatic compounds upon heating, enabling multisensory interaction. A photocurable resin was formulated using carvone, a naturally derived monoterpene from spearmint, as a nonreactive diluent, together with diurethane dimethacrylate (DUDMA) and phenylbis(2,4,6-trimethylbenzoyl)phosphine oxide (BAPO). Containing up to 25% renewable content, the formulation provides a sustainable alternative to conventional petroleum-based systems. The resulting 4D-printed materials demonstrate stable mechanical and shape-memory performance, a transition temperature of 80 °C, and effective antimicrobial activity against *E. coli* and *S. aureus*. Polymerisation kinetics in the presence of carvone were evaluated using HEMA and MMA as model monomers. HEMA reached 73% conversion within 30 min, whereas MMA achieved only 19% under identical conditions. These results indicate that carvone is more compatible with methacrylate-based systems and is therefore better suited as a nonreactive diluent for methacrylate formulations than for acrylate-based systems. By integrating olfactory responsiveness, antimicrobial functionality, and renewable feedstocks, this platform enables 4D-printed structures for healthcare, consumer products, and interactive devices.

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Interactive materials; adaptive manufacturing; unreactive diluent; shape memory; antimicrobial



## Highlights

- Biobased 4D interactive materials fabricated via adaptive manufacturing
- Carvone used as renewable unreactive diluent in sustainable photoresins
- Printed structures actively reshape and emit aroma upon thermal stimuli
- Advances sustainable additive manufacturing toward multisensory materials

## Introduction

Adaptive manufacturing is emerging as a next-generation fabrication paradigm in which printed materials exhibit real-time responsiveness and multifunctionality through the dynamic coupling of structure and function. This capability represents a significant advance in smart fabrication, enabling material properties to evolve post-printing in response to external stimuli [1]. Within this context, masked stereolithography (mSLA) has become a particularly powerful platform due to its scalability, high resolution, and compatibility with functional additives. These attributes position mSLA as an enabling technology for adaptive manufacturing, where the physical, chemical, and sensory characteristics of printed structures can be programmed and reprogrammed rather than remaining static. Such dynamic behaviour opens new opportunities for reconfigurable devices in soft robotics, biomedical engineering, and interactive systems.

Despite these advantages, the environmental footprint of photopolymer-based additive manufacturing remains a pressing concern. The production, use, and disposal of large volumes of resin contribute to pollution, greenhouse gas emissions, and resource depletion [2]. Global plastic waste reached 425 million tons in 2020, with only 22% recycled and 15% mismanaged, and by 2023, 90.4% of plastic production remained

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fossil-based [3]. These trends highlight the urgent need for sustainable materials and waste-free processing strategies in vat photopolymerization [4].

Commercial photopolymer inks typically rely on fossil-derived (meth)acrylates that form permanent thermoset networks upon UV curing. To advance sustainable additive manufacturing, significant efforts have focused on developing renewable monomers and environmentally benign resin formulations [5]. In parallel, dynamic covalent chemistry has enabled the emergence of vitrimers – cross-linked polymer networks capable of topological rearrangement through reversible bond-exchange reactions. Vitrimer-based 3D printing has leveraged diverse dynamic chemistries, including disulfide [6–8], Diels–Alder [9,10], imine [11], β-hydroxyl ester [12,13], carbamate [14–16], and boronate ester linkages [17], to achieve self-healing, shape-memory, and reprocessing capabilities [18]. Among these, carbamate-based networks are particularly attractive due to their strong hydrogen bonding, mechanical robustness, and transcarbamoylation-driven reconfigurability, which can be further enhanced by catalysts such as 2,3,4,6,7,8-hexahydro-1H-pyrimido[1,2-a]pyrimidine (TBD) [16,19].

However, the processing of vitrimer resins remains challenging. Their inherently high viscosity often necessitates the use of low-molecular-weight reactive diluents (typically acrylates) to achieve printability[1,20,21]. While effective, these reactive diluents become covalently integrated into the network during curing, limiting their ability to introduce functionalities that require chemical independence. As a result, current strategies for functionalising 3D-printed vitrimers rely heavily on modifying the covalent backbone, a synthetically demanding approach that restricts versatility and scalability.

To address this limitation, we introduce a fundamentally different strategy: the incorporation of a biobased, unreactive diluent that remains physically entrapped within the vitrimer network. Rather than participating in polymerisation, this unreactive diluent functions as a multifunctional plasticiser, simultaneously reducing viscosity to enable high-fidelity printing and imparting antimicrobial and olfactory responsiveness to the final material. This approach establishes a highly adaptable design pathway for sustainable, multifunctional 3D-printed vitrimers.

Essential oils provide a compelling platform for such functionality. These naturally derived compounds exhibit antibacterial, antifungal, insecticidal, and viscosity-reducing properties and are widely used in cosmetics, food, and healthcare [22–28]. While previous studies have explored essential-oil-derived reactive monomers for photopolymer systems [29], these approaches often suffer from limited printing fidelity or constrained functional tunability due to covalent incorporation. In contrast, the present work employs carvone, a monoterpene from spearmint and parsley, as a biobased, unreactive diluent that remains physically entrapped within a permanent vitrimer network. This design preserves carvone's intrinsic aromatic and antimicrobial properties, enabling programmable sensory functionality not achievable in reactive-monomer systems.

As materials science progresses from static 3D structures to stimuli-responsive 4D systems, integrating sensory responsiveness introduces a new dimension of interactivity. Traditional 4D materials respond to environmental cues such as temperature or light, whereas sensory-augmented systems actively influence their surroundings by engaging additional human senses [30–33]. Building on our previous demonstration of a 4D flower capable of simultaneous shape and colour changes, we now introduce olfactory responsiveness as a third sensory modality [1]. The resulting printed structures exhibit coordinated transformations in shape, colour, and scent, creating a fully immersive multisensory experience.

To realise this concept, we developed a sustainable photocurable resin composed of carvone as an unreactive diluent and diurethane dimethacrylate (DUDMA) as a cross-linker, with BAPO as the photoinitiator (Figure 1). The formulation is simple, waste-free, and compatible with mSLA printing. The printed materials demonstrate strong antimicrobial activity, shape reprogrammability, temperature-triggered shape memory, and controlled aroma release. Essential oils derived from rapidly renewable herbaceous plants further enhance sustainability while providing inherent antimicrobial and olfactory functionality.

Overall, this work introduces a new paradigm for additive manufacturing that integrates time-dependent responsiveness with sensory interactivity. By embedding olfactory cues within a stimuli-responsive vitrimer framework, we expand the functional landscape of 4D materials to include multisensory engagement. This strategy offers new opportunities for adaptive systems in healthcare, wearable technologies, consumer products, and interactive environments, and demonstrates the potential of biobased multifunctional materials to support the development of next-generation printed systems that are structurally dynamic, environmentally sustainable, and sensorially enriched.

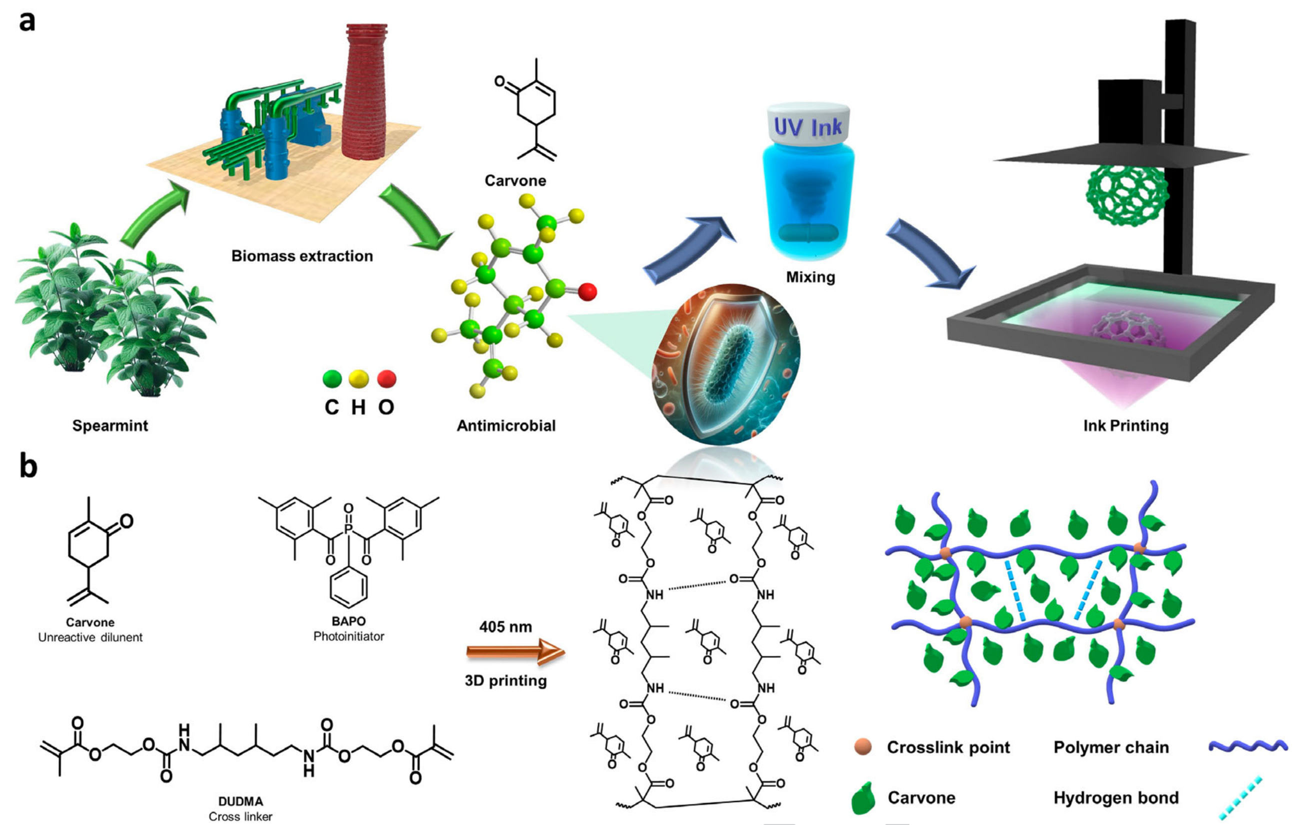




**Figure 1.** (a) Schematic for the preparation of the ink and printing process; (b) Reaction scheme and structural representation of the ink photopolymerization process.



## Methods

### *Materials*

Diurethane dimethacrylate, a mixture of isomers (DUDMA, 97%), phenylbis(2,4,6-trimethyl benzoyl) phosphine oxide (BAPO, 97%), 2-hydroxyethyl methacrylate (HEMA, 99%, containing $\leq$50 ppm monomethyl ether hydroquinone as an inhibitor), methyl methacrylate (MMA, containing $\leq$30 ppm MEHQ as an inhibitor, 99%), dimethyl sulfoxide-$d_6$ ($(CD_3)_2S=O$, 99.5 atom % D), and 2-propanol (99.5%) were purchased from Sigma?Aldrich. (R)-(-)-Carvone (99.0%) was obtained from TCI (Tokyo Chemical Industry, Japan), and 2,3,4,6,7,8-hexahydro-1H-pyrimido[1,2-a]pyrimidine (TBD, 97%) was obtained from BLD Pharmatech GmbH.

**Table 1.** Composition of the evaluated resin formulations. Each formulation contains 0.5 wt% BAPO photoinitiator, calculated relative to the combined weight of DUDMA and either HEMA or carvone. Additionally, Resin 1 and Resin 2 include 0.5 wt% TBD.

| | Monomer (% w/w) | | | |
|---|---|---|---|---|
| Sample | DUDMA | HEMA | Carvone | TBD |
| Resin 1 | 75 | 0 | 15 | Yes |
| Resin 2 | 75 | 0 | 25 | Yes |
| Resin 3 | 75 | 25 | 0 | No |
| Resin 4 | 75 | 0 | 25 | No |

### *Resin formulations*

The liquid printing precursor was prepared by mixing carvone and DUDMA at a weight ratio of 1:3. DUDMA, carvone, BAPO (0.5 wt% relative to the total resin weight), and TBD (0.5 wt% relative to the total resin weight) were magnetically stirred at room temperature for 30 min to obtain a homogeneous resin mixture (Table 1). In fact, successful printing was achieved not only with formulations containing 25% renewable components, but also with those containing 15% and 20%. However, at higher renewable content levels, particularly at 25%, the viscosity of the photocurable resin was already significantly reduced. As a result, we chose not to further increase the proportion of unreactive diluent, since doing so would clearly require prolonged curing times, which could compromise printing efficiency and fidelity. This trend is further substantiated in the subsequent discussion. Nonetheless, based on our results, we believe that achieving a renewable content of up to 30% remains a feasible target.

## Vat photopolymerization

Printing was conducted via a bottom-up 3D printer (Phrozen Sonic 4 K 2022) with a 405 nm ParaLED Matrix 3.0 light source with 2.3 mW/cm$^2$ irradiance. The print models were prepared via the Phrozen Dental Synergy Slicer. The samples containing the TBD catalyst were printed with a layer height of 100 μm and an exposure time of 30 s. The samples without the TBD catalyst were printed with an exposure time of 12 s, whereas those with HEMA as a substitute had an exposure time of 6 s. Following the printing process, the printed samples were cleansed with a paper towel moistened with isopropyl alcohol to eliminate any surface remaining resin, air-dried for 10 min and then post-UV cured (405 nm, LED radiant of 6 W) for 20 min at ambient temperature. The antimicrobial test films were prepared by casting liquid resin into PS Petri dishes and curing them in a post-UV box for 50 s. The printing parameters (pull-up distance, speed, waiting time, transition layers, burn-in layers, and normal exposure time) were selected based on the default settings recommended by the printer manufacturer for standard photopolymer resins; no additional optimisation was performed beyond these preset values.

## Characterisation

$^1$H NMR spectroscopy of carvone and photopolymerization kinetics were performed with a Bruker Ascend™ NMR600 spectrometer at 25 °C in DMSO-$d_6$ via 16 scans. For the conversion calculation, the one using MMA as model, we use protons intensity from propenyl hydrogens at 4.72–4.81 ppm as internal standard, using the proton intensity of C = C from methacrylate at 5.65–6.05 ppm as the base. Conversion yield is calculated by comparing the time-dependent decrease in C = C proton intensity to the internal standard. All NMR kinetic results were calculated in triplicate. For conversion calculations using MMA as a model compound, the proton intensity of the propenyl hydrogens (δ 4.72–4.81 ppm) is employed as an internal standard, while the proton intensity of the methacrylate C = C bonds (δ 5.65–6.05 ppm) serves as the reference. By monitoring the decrease in the C = C proton signal over time and normalising to the initial intensity, the conversion yield is determined. During the process, we observed that the proton signal from the enone (δ 6.82–6.88 ppm) remained constant, indicating that the enone group is chemically inert under the UV polymerisation conditions. Therefore, for conversion calculations using HEMA, the same procedure is followed, but the unchanging enone signal is employed as the internal standard.

Attenuated total reflection-Fourier transform infrared (ATR-FTIR) spectra were acquired via a Bruker VERTEX 70 spectrometer equipped with a diamond single-reflection ATR accessory and scanned over the 4000–700 cm$^{-1}$ range. To further investigate the absorbance of the dynamic changes, an ATR accessory with a heater was used to record the spectra across a temperature range of 40 °C to 120 °C.

Tensile testing was performed in accordance with a smaller size specimen for ISO 527-2:1996 on an Instron testing apparatus equipped with a load cell capacity of 1 kN to ascertain the mechanical characteristics of the materials under examination. The strain rate was maintained at 5 mm/min, and the samples for tensile testing were printed in a dumbbell configuration with dimensions of 48.75 mm × 3.25 mm × 1.3 mm. The measurement was repeated three times to ensure reproducibility. All tensile specimens were tested in the as-printed condition, prior to any thermal treatment.

The thermal stability of the cured resins was analyzed via thermogravimetric analysis (TGA) on a TA Instruments Q series with an autosampler using a temperature ramp from 25 to 700 °C at a rate of 10 °C min$^{-1}$ under atmospheric and nitrogen flows. Furthermore, the thermal volatilisation of carvone at 60, 80, and 120 °C was also analyzed via TGA (40 °C min$^{-1}$).

The glass transition temperature ($T_g$) was measured via a TA Instruments Discovery HR20 instrument in DMA mode (1 Hz, 0.2% strain, heating rate: 3 °C min$^{-1}$). The test samples had dimensions of 20 mm × 5 mm × 2 mm.

Control and UV-cured films were prepared to evaluate their antibacterial efficacy. Antimicrobial tests were conducted using Escherichia coli (*E. coli*) and Staphylococcus aureus (*S. aureus*) as model microorganisms, following the national standard GB/T 20944.3-2008 and protocols based on previously reported methods [34, 35].

## Results and Discussion

To investigate methacrylate reactivity via photopolymerization in carvone, we selected two representative monomers, HEMA and MMA, to observe their free radical polymerisation at room temperature under UV excitation (405 nm, LED radiant power of 6 W) for 0–30 min. The molar ratio of monomethacrylate to carvone was maintained at 1:1, with 0.5 wt% BAPO as the photoinitiator. Prior to this, a control experiment was performed to rule out potential interference from carvone side reactions (Fig. S1). As shown in Figure 2a and 2b, HEMA exhibited good reactivity, achieving a

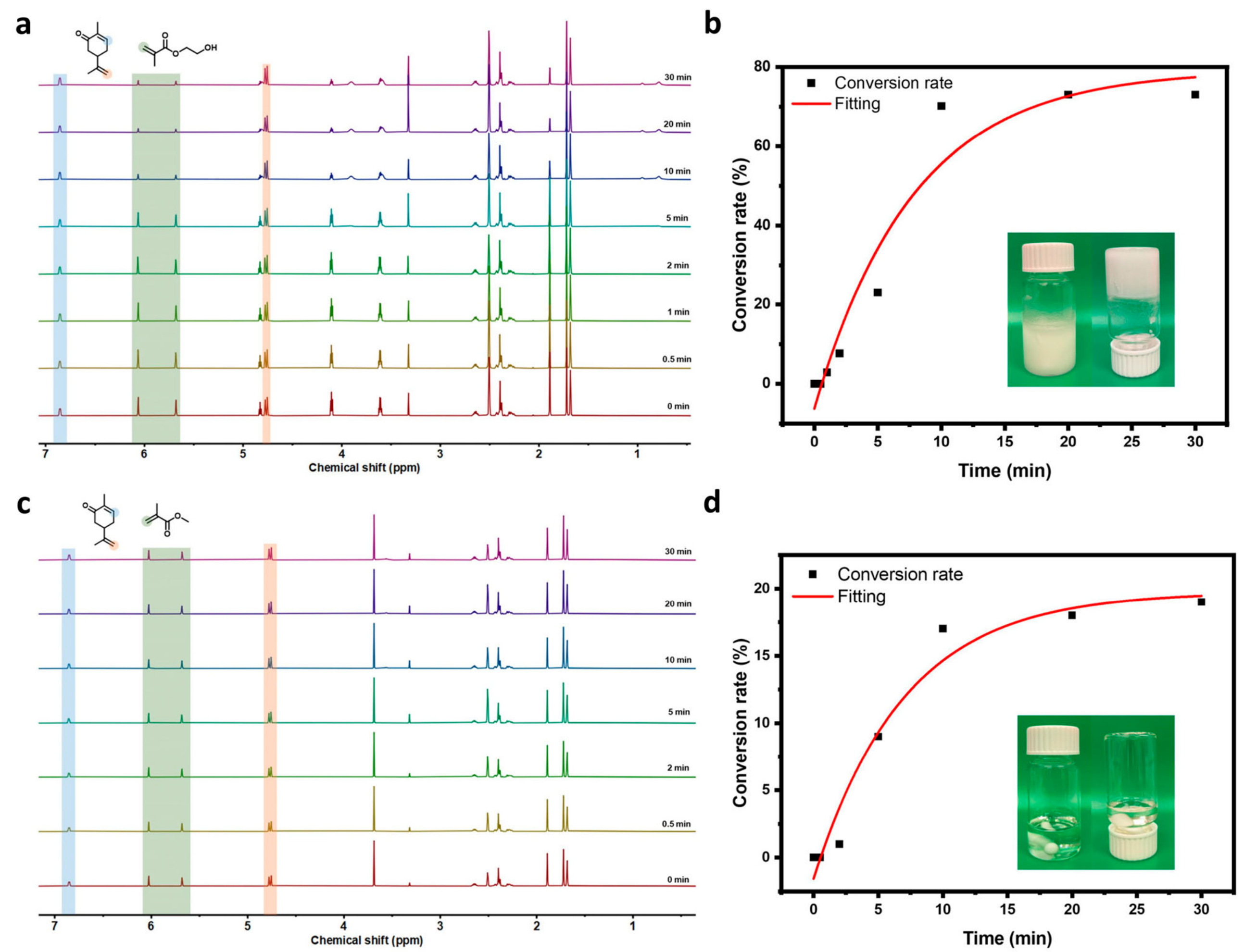




**Figure 2.** (a) $^{1}$H NMR spectra of HEMA polymerisation via radicals mediated by increasing UV radiation times in carvone; (b) Plot of the conversion rate of unsaturated bonds over irradiation time monitored by $^{1}$H NMR spectroscopy; (c) $^{1}$H NMR spectra of MMA polymerisation via radicals mediated by increasing UV radiation times in carvone; (d) Plot of the conversion rate of unsaturated bonds over irradiation time monitored by $^{1}$H NMR spectroscopy.

73% conversion rate within 30 min (see Supporting Information, Table S1). During the polymerisation process, the solution became increasingly cloudy and eventually solidified, indicating that the polymerised product of HEMA is insoluble in carvone. Conversely, MMA showed poor reactivity, with only a 19% conversion rate over the same time period (see Supporting Information, Table S2). Throughout the polymerisation process, the solution remained clear and transparent, suggesting that the polymerised product of MMA is soluble in carvone. It should be noted that the differing solubility of poly(MMA) and poly(HEMA) in carvone creates distinct polymerisation environments (homogeneous vs. heterogeneous), which precludes a direct comparison of intrinsic reactivity. Rather, this experiment was designed to evaluate the compatibility of carvone with acrylate versus methacrylate systems. The results clearly demonstrate that carvone is significantly more compatible with methacrylate-based formulations, as evidenced by the higher conversion achieved and the absence of macroscopic phase separation. This finding supports the rationale for employing carvone as an unreactive diluent specifically in methacrylate systems. Furthermore, the incomplete conversion observed even under prolonged UV irradiation indicates that full consumption of unsaturated bonds is unlikely to be achieved under the present conditions.

Subsequent to the polymerisation process, we proceeded with 3D printing using resin 2, followed by infrared (IR) spectroscopy analysis. The IR spectrum of carvone revealed characteristic absorption peaks, including the conjugated carbonyl group at 1670 $cm^{-1}$ and the stretching vibration of propenyl hydrogens (C = C) at 1638 $cm^{-1}$ (Figs. S2a and S2b). Fig. S2b shows a gradual decline in these characteristic peaks (1670 and 1638 $cm^{-1}$) after both the post-UV and natural curing processes. The significant decrease in the 1670 $cm^{-1}$ peak is attributed to the enone stretching vibration, which is likely accompanied by evaporation into the ambient environment. Similarly, the reduction in absorbance at 1638 $cm^{-1}$, corresponding to the C = C stretch, indicates the retention of unreacted methacrylate groups from DUDMA in the final prints. To further validate these observations, variable-temperature ATR-FTIR and TGA analyses were employed to monitor carvone volatilisation from the prints between 40 and 120 °C. Additionally, the effects of TBD on chemical structure variations at elevated temperatures were investigated via the use of resin 3 for printing. The ATR-FTIR results (Figs. S3 and 3a) revealed no significant structural changes. Figure 3a shows that the absorbance at 1670 $cm^{-1}$ increased with increasing number of heating and cooling cycles. Additionally, direct measurements of liquid carvone evaporation rates corroborated the IR findings (Figure 3b). The 120 °C condition was intentionally selected to accelerate evaporation and to serve as a preliminary assessment of aging behaviour. Although the possibility of thermally induced side reactions (e.g. oxidation or dimerisation) of carvone at 120 °C cannot be ruled out, the key finding is that carvone-derived olfactory functionality persists even after exposure to 120 °C for 2 h, indicating that a sufficient quantity of carvone remains entrapped within the polymethacrylate network to enable long-lasting aroma release. The evaporation rate was strongly temperature dependent: at 40 °C, approximately 87% of the carvone weight remained after 1 h, whereas at 120 °C, the carvone evaporated completely within 5 min. To investigate whether elevated temperatures would cause carvone to escape from the polymer network and induce cracking, printed films were subjected to heating at 120 °C for 2 h. Remarkably, the films remained intact and exhibited no macroscopic contraction detected. Characteristic absorbances were measured at different intervals to assess chemical stability. As shown in Figure 3c, the IR spectra at 120 °C for 0.5 and 2 h were nearly identical, with no carvone residue detected on the surface. Interestingly, upon further increasing the temperature, the absorbance at 1670 $cm^{-1}$ intensified again (Figure 3d). This increase was attributed to compressive forces from the ATR and prolonged heating, which caused residual carvone trapped within the polymer network to migrate to the surface of the film. The TGA data indicate a non-negligible mass loss of approximately 2–3% over 60 min at 80 °C (Figure 3b). This gradual loss of the carvone plasticiser has important implications for long-term shape memory cycling stability. At elevated temperatures under external mechanical force, carvone migration is accelerated and the softened polymer becomes increasingly susceptible to fracture. These findings highlight that the gradual loss of the carvone plasticiser upon thermal cycling is the primary factor governing long-term shape memory performance. To address this limitation, implementing strategies like microencapsulation represents a highly promising avenue for prolonging plasticiser retention.

mSLA printing is a process in which a liquid photosensitive resin is cured layer by layer using UV light to create 3D geometrical shapes. Investigating the layer curing process provides valuable insights into the kinetics of DUDMA photopolymerization. The absorption of UV light by liquid photopolymer resins generally follows the Beer–Lambert law, expressed as Cd = Dp*ln(E/Ec). By evaluating a series of curing depths (Cd) and exposure doses (Ed), the penetration depth (Dp) and critical energy (Ec) of the resin can be determined. Dp indicates the resin's sensitivity to variations in light intensity. Ec defines the minimum energy needed to create a solid layer and is thus the threshold energy for successful printing. As shown in Figure 4a and 4b, the values varied significantly among the different resins. Compared with the HEMA-containing resin (Resin 3, Table 1), the carvone-containing resin (Resin 4, Table 1) requires more energy for solidification. This difference is attributable to carvone's unreactivity and the reduced concentration of methacrylate groups in Resin 4. Observed from Fig. S4, the addition of 0.5 wt% TBD to the carvone-containing resin (Resin 2, Table 1) further increases the required energy. This observation suggests that TBD influences the free-radical polymerisation process, consistent with previous studies reporting that TBD can affect polymerisation kinetics by retarding the initial polymerisation rate[36, 37]. Additionally, we observed distinct curing behaviours: Resin 4 formed a jelly like gel, whereas Resin 3 produced a compact, stronger layer. However, owing to the good penetration depth of Resin 4, the layer-by-layer printing process was minimally affected. The transparency of the resin allows the previously printed layer to undergo further curing during the current layer's exposure, thereby maintaining sufficient strength to form an intact 3D structure. To compare the mechanical properties of these resins, tensile stress–strain tests were conducted. Commercial 3D printing resins typically contain multiple proprietary components (including viscosity modifiers, defoaming agents, and other undisclosed additives); therefore, to avoid confounding effects from unknown additives and to ensure a controlled and interpretable comparison, the carvone-based formulation was benchmarked against a HEMA-containing resin of explicitly known composition. Figure 4c and 4d, together with

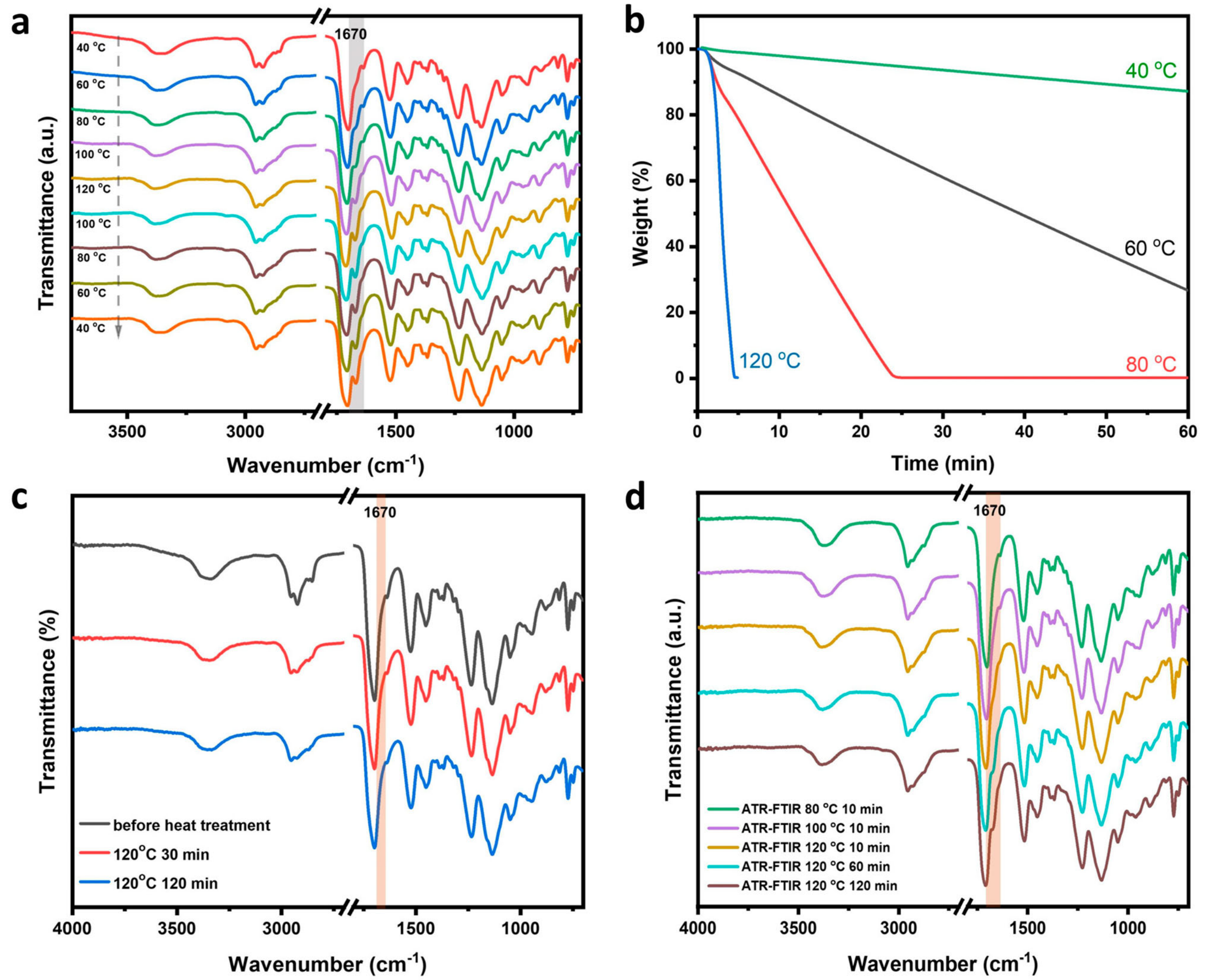




**Figure 3.** (a) ATR-FTIR spectra of the printed thin film without TBD recorded during a heating and cooling cycle at temperatures ranging from 40 °C to 120 °C; (b) Weight loss curves of carvone at 40 °C, 60, 80, and 120 °C versus time; (c) ATR-FTIR spectra of the film following exposure to 120 °C heat treatment; (d) Transmittance changes at 1670 $cm^{-1}$ after heat treatment within the 80–120 °C range.

Fig. S5 show that although the carvone-containing resin results in a lower ultimate stress than the HEMA-containing resin does, it results in greater elongation at break, mainly due to both the plasticising effect of carvone and the lower crosslink density. This finding indicates that Resin 4 offers a less tough but more flexible alternative for 3D-printed structures. The substantially lower critical energy (Ec) of the HEMA-containing resin compared with the carvone-containing resin reflects a fundamental shift in photopolymerization kinetics and network architecture. Replacing the reactive diluent (HEMA) with a non-reactive diluent (carvone) alters the polymerisation pathway: carvone reduces viscosity and enhances radical mobility but does not participate in crosslinking, thereby delaying vitrification and enabling network formation at lower doses. However, its non-reactive nature yields a final network with lower effective crosslink density, higher free volume, and enhanced plasticisation, characteristics that manifest as reduced modulus and lower $T_g$ relative to the HEMA-based counterpart. In contrast, HEMA becomes covalently incorporated, increasing crosslink density and promoting earlier vitrification, which raises Ec but produces a mechanically stronger and more dimensionally stable network. Crosslink density calculations based on DMA data confirmed that the HEMA-containing formulation exhibits a higher crosslink density than the carvone-containing counterpart (see Supporting Information, Figure S6 and Table S3). Furthermore, the desktop printer employed in this study operates at a relatively low light intensity of 2.3 mW/$cm^2$. The prolonged exposure time observed for the TBD-containing resin is

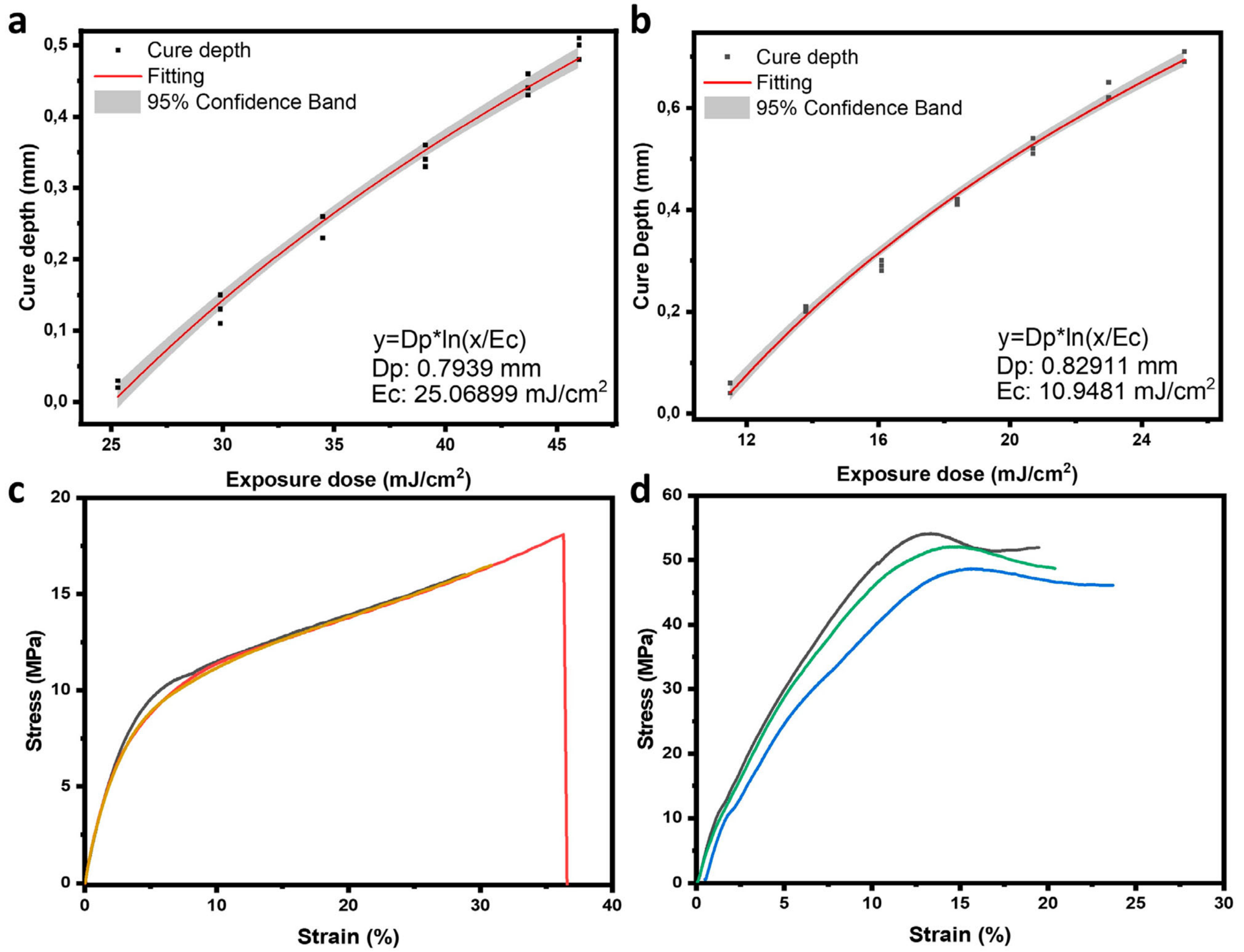


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**Figure 4.** Photo flux versus polymerisation depth. (a) Resin 4; (b) Resin 3; (c) Tensile stress–strain curve of a film printed from resin 4 (colour lines represent three repetitions); (d) Tensile stress–strain curve of a film printed from resin 3 (colour lines represent three repetitions).

attributable to both TBD-induced chemical interference with radical photopolymerization, as reported in previous studies [36, 37], and the modest light intensity of the printer.

Resin 2 (containing 25 wt% carvone and 0.5 wt% TBD) was employed for all subsequent mechanical and shape-memory characterisations. As discussed earlier, the unreactive carvone serves as a more flexible diluent, promoting shape reconfiguration and memory capabilities by reducing the rigidity of the cross-linked urethane network under low force. Resin 2 was subsequently selected for further printing because the literature suggests that TBD can enhance shape reprogramming and memory properties [16, 19]. As shown in Figure 5a, a flawless elephant model was successfully printed. To demonstrate the shape reprogramming capabilities of the ink formulation, a dumbbell and a hollow lantern were printed and then reshaped under external force at 80 °C. The objects were transformed into different geometric forms, as shown in Figure 5b and 5c. In Figure 5b, the thin film undergoes significant shape transformation, with the original printed shape reprogrammed into distinct configurations (e.g. spiral and compressed forms) through multiple cycles of heating and reshaping. This reprogramming ability highlights the material's capacity to maintain structural integrity while adopting new geometries. Similarly, Figure 5c shows the adaptability of the hollow lantern, which was reprogrammed into various shapes, such as diamond, crown, and peanut forms. These findings underscore the material's dynamic reconfigurability, driven by external forces applied at elevated temperatures, enabling the creation of complex and varied shapes from a single printed object.

Materials with shape memory provide promising opportunities to bridge the gap between thermosets and thermoplastics, facilitating the creation of 3D printing materials with highly customizable geometries

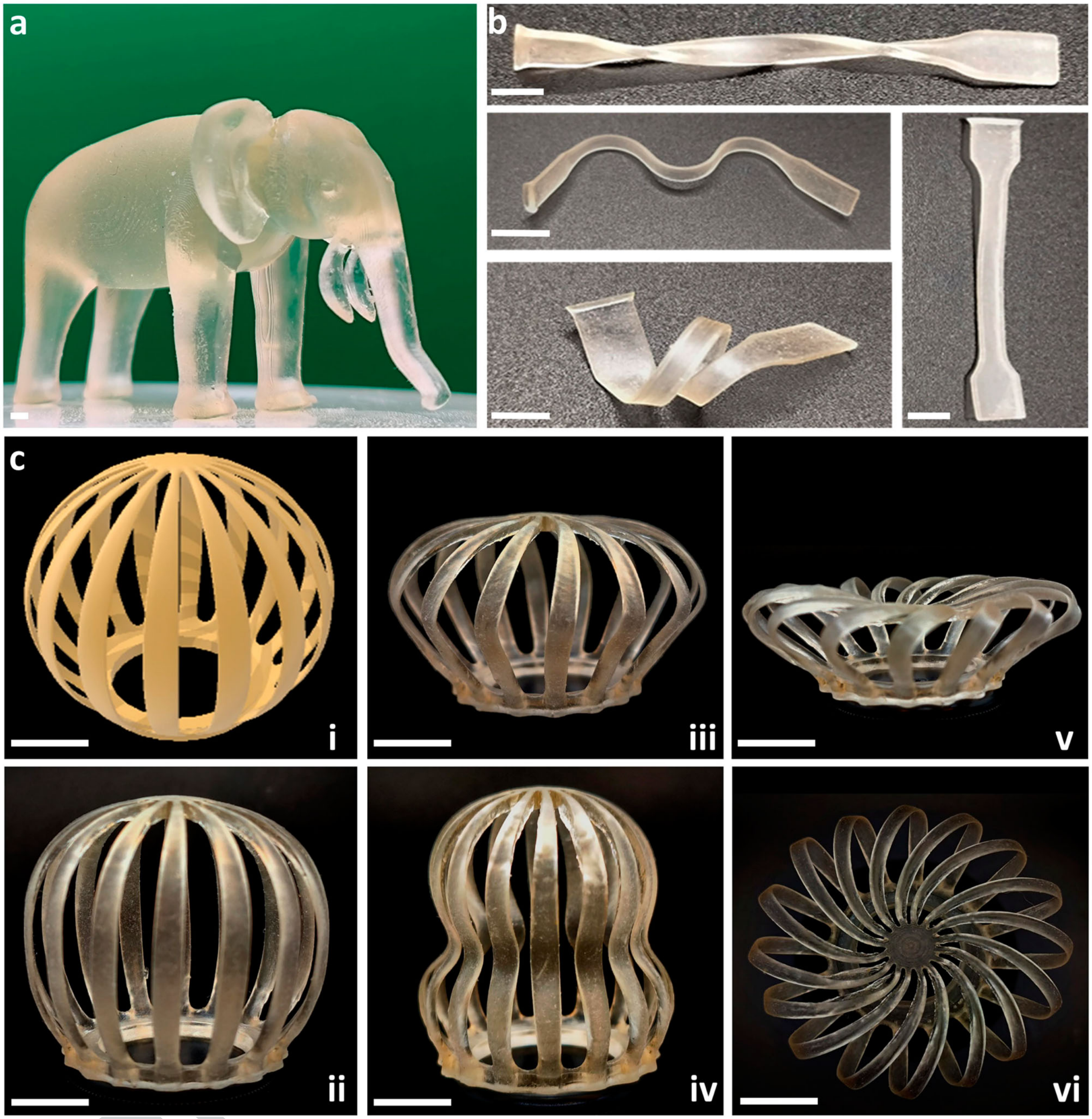


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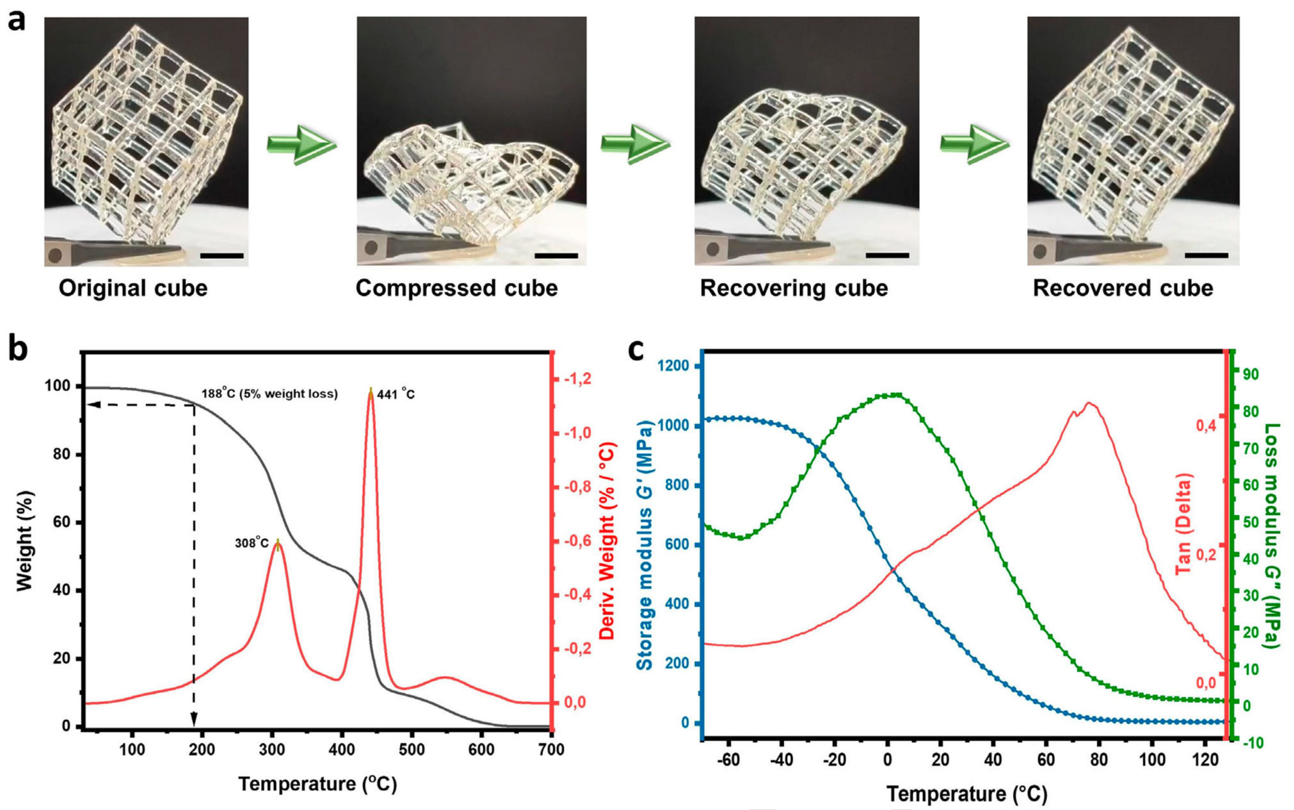
**Figure 5.** (a) Digital photos of the printed elephant model; (b) Digital photo of the reprogramming shape of the dumbbell-style spline; scale bar, 6 mm; (c) The 3D hollow lantern model, its printed prototype, and the shape following a series of reprogramming stages. The sequence includes: (i) the digital model of the hollow lantern, (ii) the initial 3D-printed lantern prototype, (iii) deformation into a diamond-shaped structure, (iv) further compression into a peanut-like shape, (v) reconfiguration into a crown-like form, and (vi) the top-down view of the final crown shape. Scale bar: 6 mm.

tailored to specific needs. As shown in Figure 6a and the supplementary video, the printed cube becomes malleable at 80 °C, allowing its shape to be manipulated during the shape memory transition, resulting in a compressed cube. Upon reheating, the compressed cube restores its original shape. During this process, carvone volatilisation emits a specific aroma. Despite this volatilisation, the cube structure remains intact without cracking, demonstrating the potential of carvone for developing advanced multisensory 4D materials. Figure 6b further shows the exceptional thermal resistance of the material, with only a 5% weight loss at 180 °C, which supports multiple cycles of shape manipulation. Additionally, the observed glass transition temperature ($T_g$) peak (Figure 6c) confirms that shape manipulation occurs at relatively low temperatures.




Colour online, B/W in print

**Figure 6.** (a) Printed cube deformed at 80 °C and corresponding shape memory behaviours at 80 °C; scale bar, 5 mm; (b) TGA and DTG curves of the printed film; (c) Storage modulus, storage modulus, and loss factor curves for a printing film.



The shape memory behaviour in this system is primarily governed by the elastic shape-memory effect characteristic of polyurethane acrylates, wherein hard segments and intermolecular hydrogen bonding serve as physical netpoints to stabilise the temporary shape. Upon reheating, chain mobility and entropic elasticity are restored, and the thermally induced disruption of hydrogen bonds facilitates recovery to the original configuration. It should be noted that transcarbamoylation catalyzed by TBD is not the dominant shape memory mechanism in this system, as the polymer formulated without TBD also exhibits satisfactory shape memory performance [15]. Attempts to quantitatively characterise cyclic shape memory performance via DMA were unsuccessful because the carvone-containing samples fractured during repeated stretching at 80 °C. The intensified thermal conditions accelerated carvone escape from the printed spline, leading to crack formation and premature failure. Nevertheless, manual shape memory cycling was successfully demonstrated for more than six cycles, albeit with progressive stiffening of the material due to the gradual release of carvone.

In certain applications, the antimicrobial and antifungal properties of 3D-printed materials are highly desirable, particularly for medical devices requiring strict hygiene. Staphylococcus aureus (*S. aureus*) and Escherichia coli (*E. coli*) are commonly used bacterial species to evaluate antimicrobial properties. Here, we employed these two bacterial strains to assess the antimicrobial efficacy of the films. All the films were fabricated from their respective resins (e.g. Film 1 from Resin 1). As shown in Figure 7a, Film 1 and Film 2 (containing 15 wt% carvone and 25 wt% carvone, respectively) exhibited superior antimicrobial activity against *S. aureus* compared with Film 3 (containing 25 wt% HEMA). However, for *E. coli*, Film 1 demonstrated lower antimicrobial activity than Film 2 but outperformed Film 3. This difference can be attributed to the reduced weight percentage of carvone in Film 1. To evaluate the long-term performance, the antimicrobial properties were reassessed after two months. The results (Figure 7b) were consistent with the initial findings, showing that films containing carvone retained desirable antimicrobial activity against both *S. aureus* and *E. coli* over time. Carvone, as a naturally occurring monoterpene essential oil, possesses well-documented intrinsic antimicrobial properties, the mechanisms of which have been extensively characterised in the literature [26–28]. As a free-state plasticiser, carvone is prone to gradual migration toward the polymer surface.

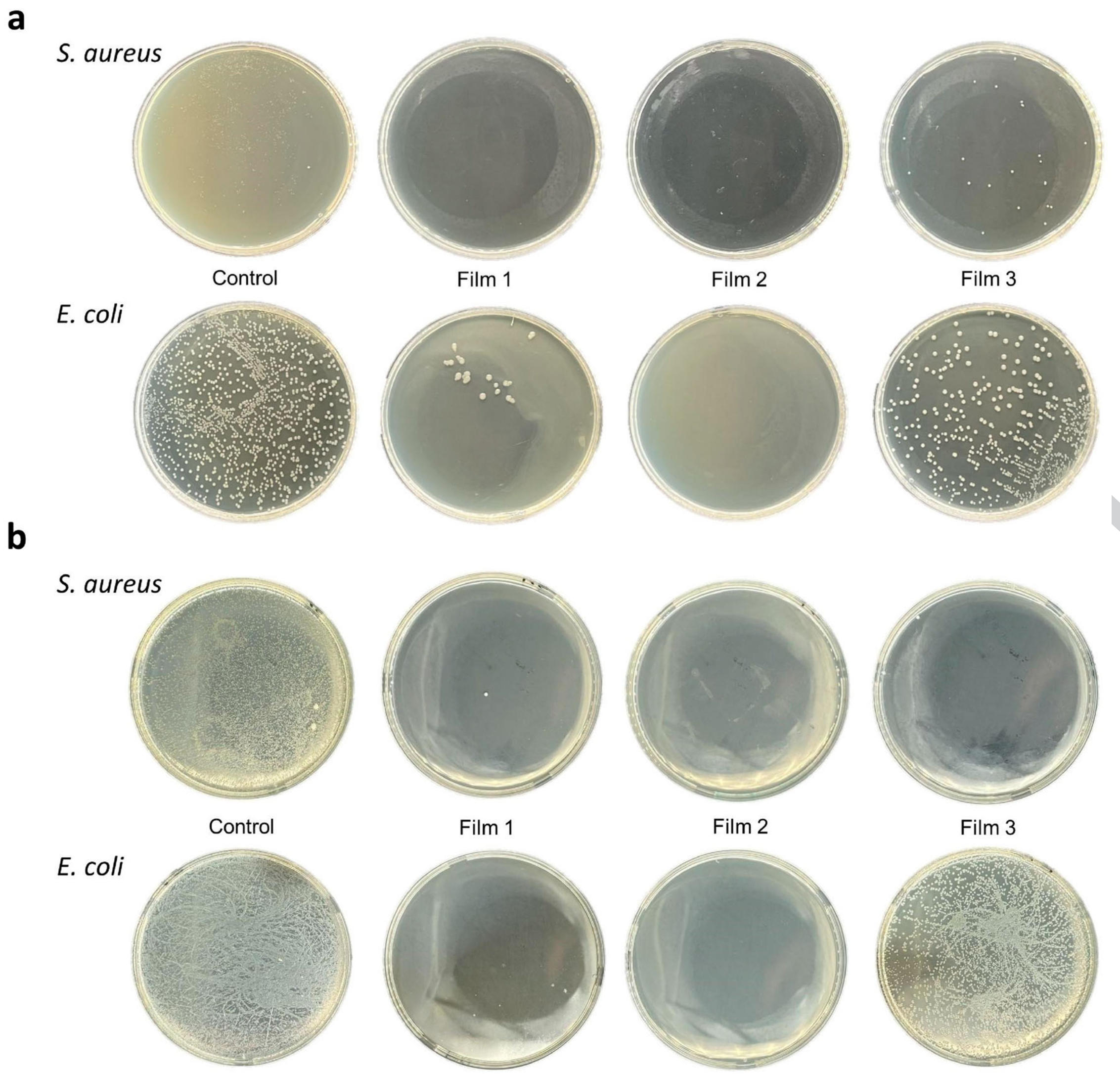





**Figure 7.** UV ink formulation for the antimicrobial test of cured films with *S. aureus* and *E. coli* bacteria. (a) First test; (b) The test after 2 months.

Thermal activation accelerates this migration, intensifying aroma release to readily detectable levels; at room temperature, the release is more subdued and perceptible only at close proximity. It should be noted that the antimicrobial assays were performed on cast films rather than on 3D-printed objects with representative surface finish and geometry, and the potential influence of printing-induced surface topography was not captured. Given that the 25 wt% carvone content can migrate to the surface, comparable antimicrobial efficacy is anticipated on printed parts; Additionally, micro-scale surface porosity or heterogeneity resulting from carvone migration, although not evident as macroscopic porosity on sample surfaces, cannot be entirely ruled out.

## Conclusions

In this work, we successfully developed and characterised a sustainable photocurable ink formulation for adaptive manufacturing, employing carvone as a non-reactive diluent and diurethane dimethacrylate (DUDMA) as the cross-linker. The incorporation of carvone imparted notable antibacterial activity against *S. aureus* and *E. coli*, attributable to its inherent antimicrobial properties, while also enhancing the flexibility and softness of the printed materials. These characteristics enabled shape reconfigurability, facilitating complex geometric transformations in the post-printing state. A representative printed cube exhibited thermally induced malleability at 80 °C, allowing for controlled deformation during the shape memory cycle and full recovery of its original form upon reheating. Notably, despite partial volatilisation of carvone at elevated temperatures, the printed structures retained their structural integrity, showing no signs of cracking or structural failure. The combination of dynamic responsiveness, antibacterial performance, aromatic functionality, and shape adaptability positions these carvone-based formulations as a new class of 4D interactive materials, capable of integrating multifunctionality and environmental responsiveness within additive manufacturing platforms. These results highlight the potential of this system for the development of advanced antimicrobial materials, with promising applications in medical devices, interactive prototypes, and custom-tailored components. The gradual loss of the carvone plasticiser upon thermal cycling represents a critical factor governing long-term shape memory cycling stability, and strategies such as microencapsulation represent promising avenues for prolonging carvone retention.

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Q2 Project administration.

## Author contributions

CRediT: **Tao Zhang:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing; **Kunlin Chen:** Conceptualization, Methodology, Writing – review & editing; **Kylian Janssen:** Formal analysis, Validation, Writing – review & editing; **Vincent S. D. Voet:** Conceptualization, Methodology, Supervision, Writing – review & editing; **Rudy Folkersma:** Conceptualization, Methodology, Supervision, Writing – review & editing; **Katja Loos:** Conceptualization, Methodology, Project administration, Supervision, Writing – review & editing.

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| Fig. 1 | **Alt-Text:** Two diagrams showing biomass extraction from spearmint into carvone based ultraviolet ink and a photopolymer network for antimicrobial 3D printing. |
| Fig. 1 | **Extended Description:** The figure shows two related diagrams of a carvone based ultraviolet curable ink for antimicrobial 3D printing. The upper illustration labeled a begins with a spearmint plant on the left, followed by an industrial biomass extraction facility. A small key with spheres labeled C, H, and O indicates carbon, hydrogen, and oxygen atoms. Arrows lead to a structural drawing of carvone and a ball and stick molecular model annotated as antimicrobial. Another arrow points to a blue container labeled ultraviolet ink with a mixing arrow, then to a schematic of a vat type 3D printer producing a lattice object from a liquid bath marked ink printing. The lower illustration labeled b presents three separate chemical structures for carvone, the photoinitiator BAPO, and the cross linker diurethane dimethacrylate, each with text labels describing their roles. An arrow labeled 405 nanometer 3D printing points to a larger polymer network structure showing repeated urethane and methacrylate motifs. To the right, a simplified cartoon highlights polymer chains, crosslink points, dispersed carvone units, and dashed lines representing hydrogen bonds, with a key identifying each feature. |
| Fig. 2 | **Alt-Text:** Four visuals: two one dimensional nuclear magnetic resonance spectra plots and two line graphs showing conversion rate versus time trends. |
| Fig. 2 | **Extended Description:** The figure shows four visuals comparing reactions of hydroxyethyl methacrylate and methyl methacrylate with carvone under ultraviolet irradiation. The first visual is a stack of proton nuclear magnetic resonance spectra for hydroxyethyl methacrylate at irradiation times labeled 0, 0.5, 1, 2, 3, 5, 10, 20, and 30 minutes. The horizontal axis is chemical shift in parts per million, ranging from 7 to 1 with tick marks every 1 unit. Several regions are shaded to highlight peaks that decrease as time increases. The second visual is a line graph of conversion rate versus time for hydroxyethyl methacrylate. The horizontal axis is time in minutes from 0 to 30 with ticks at 5 minute intervals. The vertical axis is conversion rate in percent from 0 to 80 with ticks every 20. Black squares show data points that rise quickly then level off, and a red curve overlays the points. An inset shows two small vials, one cloudy and one appearing solid. The third visual repeats the stacked spectra layout for methyl methacrylate with the same time points and axis settings; highlighted peaks also diminish with time. The fourth visual is a line graph of conversion rate versus time for methyl methacrylate, with the horizontal axis from 0 to 30 minutes and the vertical axis from 0 to 20 percent at 5 percent intervals. Data points and a fitted curve rise gradually and then plateau. An inset shows two vials that remain clear. All data are approximate. |
| Fig. 3 | **Alt-Text:** Four line graphs showing ATR Fourier transform infrared spectra and carvone weight loss curves versus temperature, time, and wavenumber. |
| Fig. 3 | **Extended Description:** The figure shows four graphs labeled a, b, c, and d that depict variable temperature infrared spectra and mass loss of carvone. Graph a is a line graph of transmittance versus wavenumber for a printed thin film measured during a heating and cooling cycle between 40 degree Celsius and 120 degree Celsius. The x axis shows wavenumber from about 4000 to 650 inverse centimeters with a break near the center, and the y axis shows transmittance in arbitrary units. Multiple stacked spectra are labeled with temperatures, and the region at 1670 inverse centimeters is highlighted. Graph b is a line graph of carvone weight percent versus time from 0 to 60 minutes. The y axis runs from 0 to 100 percent. Four lines labeled 40 degree Celsius, 60 degree Celsius, 80 degree Celsius, and 120 degree Celsius start near 100 percent at time 0 and decrease over time, with faster loss at higher temperature. Graph c is a line graph of transmittance |

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| | percent versus wavenumber from 4000 to about 650 inverse centimeters. Three spectra are labeled before heat treatment, 120 degree Celsius 30 minutes, and 120 degree Celsius 120 minutes, with the 1670 inverse centimeters region marked. Graph d is a line graph of transmittance in arbitrary units versus wavenumber from 4000 to about 650 inverse centimeters for several conditions from 80 degree Celsius 10 minutes to 120 degree Celsius 120 minutes, again emphasizing 1670 inverse centimeters. All data are approximate. |
| Fig. 4 | **Alt-Text:** Four line graphs showing cure depth versus exposure dose for Resin 4 and Resin 3 and stress versus strain for printed films. |
| Fig. 4 | **Extended Description:** The figure shows four line graphs arranged in a two by two grid. The top left graph, labeled a, plots cure depth in millimeters on the vertical axis from 0 to 0.5 with tick marks every 0.1, against exposure dose in millijoules per square centimeter on the horizontal axis from 25 to 45 with tick marks every 5. Small square markers rise smoothly upward, and a fitted curve with a narrow confidence band follows the same increasing trend. Text inside the graph states y equals Dp times ln ( x divided by Ec ), Dp equals 0.7939 millimeters, Ec equals 25.06899 millijoules per square centimeter. The top right graph, labeled b, has the same layout, with cure depth from 0 to 0.6 and exposure dose from 12 to 24. Data points and a fitted curve again increase with dose. Text reads y equals Dp times ln ( x divided by Ec ), Dp equals 0.82911 millimeters, Ec equals 10.9481 millijoules per square centimeter. The bottom left graph, labeled c, shows stress in megapascal from 0 to about 18 versus strain in percent from 0 to 40. Three nearly overlapping lines rise steeply then level before one extends to higher strain. The bottom right graph, labeled d, plots stress from 0 to about 55 versus strain from 0 to 25. Three separate curves rise to peaks between about 30 and 50 megapascal then gradually decline. All data are approximate. |
| Fig. 5 | **Alt-Text:** Three photos showing a printed elephant model, a reprogrammable dumbbell style strip, and a hollow lantern evolving into 6 shapes. |
| Fig. 5 | **Extended Description:** The figure shows three groups of photos presenting 3D printed objects made from shape reprogrammable resin. The first photo, labeled a, shows a small translucent elephant sculpture standing on a flat surface, with a scale bar indicating about 6 millimeters. The second group, labeled b, contains four narrow dumbbell style strips laid on a dark background. One strip is straight, one is twisted, one forms a smooth wave, and one is folded into a compact zigzag, each with a 6 millimeter scale bar nearby. The third group, labeled c, displays a hollow lantern structure that changes geometry through six stages. Image i shows the initial digital style lantern with evenly spaced vertical ribs forming a closed sphere above a circular base ring. Image ii shows the printed lantern prototype as an open cage with vertical ribs. Image iii shows a compressed crown like form where ribs flare outward at the top. Image iv shows a tall peanut like shape with the ribs bowed inward at mid height. Image v shows a low, wide crown like configuration viewed from the side. Image vi shows a top down view of the final crown, where the ribs radiate outward in a circular pattern. Each lantern stage includes a 6 millimeter scale bar. |
| Fig. 6 | **Alt-Text:** Three graphs and four photos showing a printed carvone cube deforming at 80 degree Celsius and thermogravimetric and dynamic mechanical trends. |
| Fig. 6 | **Extended Description:** The figure shows four photos above two line graphs and one combined thermogravimetric and derivative weight graph. The first photo displays an open lattice printed cube resting on one edge. A scale bar of 5 millimeters lies at the bottom. The second photo shows the same cube compressed and flattened. The third photo shows the distorted cube partially returning toward its initial shape. The fourth photo shows the cube restored to an upright lattice |

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| | form, again with a 5 millimeter scale bar. Arrows between the photos indicate the sequence from original to compressed, recovering, and recovered cube at 80 degree Celsius. Below, the left graph plots weight percentage and derivative weight of a printed film versus temperature in degree Celsius. The horizontal axis ranges from 0 to 700 with tick marks every 100. Weight percentage on the left vertical axis ranges from 0 to 100 with tick marks every 20 and decreases gradually, with a marked point at 188 degree Celsius indicating 5 percent weight loss. The derivative weight on the right vertical axis ranges approximately from 0 to about negative 1.2 percent per degree Celsius, with peaks labeled near 308 degree Celsius and 441 degree Celsius. All data are approximate. The right graph shows dynamic mechanical analysis of a printing film, with temperature on the horizontal axis from about negative 60 to 120 degree Celsius. The left vertical axis shows storage modulus in megapascals from 0 to 1200 with tick marks every 200. One storage modulus curve starts high at low temperature and decreases as temperature increases. A second modulus or related curve starts lower and forms a broad peak at intermediate temperature before declining. The right vertical axis shows the loss factor Tan Delta from 0.0 to about 0.4 and loss modulus from about negative 10 to 90 megapascals. The Tan Delta curve rises to a distinct peak around the glass transition region and then falls as temperature increases. All data are approximate. |
| Fig. 7 | **Alt-Text:** Two photos showing S. aureus and E. coli colonies on control, Film 1, Film 2, and Film 3 agar plates at the first test and after 2 months. |
| Fig. 7 | **Extended Description:** The figure shows two sets of agar plate photos used to compare bacterial growth on different ultraviolet cured films. The upper set, labeled a, displays Staphylococcus aureus in the top row and Escherichia coli in the bottom row. For each bacterium, 4 circular plates are arranged from left to right and labeled below as Control, Film 1, Film 2, and Film 3. The Control plates show dense, uniform bacterial coverage. The plates for Film 1 and Film 2 show mostly clear agar with very few visible colonies. The plates for Film 3 show more numerous, scattered colonies than Film 1 and Film 2 but less growth than the Control. The lower set, labeled b, repeats the same layout for a second assay performed after a 2 month interval, again with Staphylococcus aureus in the top row and Escherichia coli in the bottom row. The Control plates still show heavy growth. The plates for Film 1 and Film 2 remain largely clear with sparse colonies. The plates for Film 3 display intermediate growth, with colonies more evident than on Film 1 and Film 2. No numeric scales or axis labels are present. |